LABPUR SMART SPATIAL DECISION SUPPORT SYSTEM

Scientific Geospatial Assessment for Disaster Management & Block Development

Project Lead & Researcher: MANGAL MONDAL

Target Authority: Office of the BDO, Labpur

Engine: Mobile Python GIS (Pydroid 3)

1. Executive Summary

Conceptualized and engineered by **Mangal Mondal**, this data-backed system addresses monsoon flood vulnerabilities along the Kuye river basin and evaluates public health accessibility. Using spatial mathematics, it calculates precise priority indices to eliminate bias in local resource allocation.

2. Key Deliverables & System Architecture

- **Haversine Proximity Calculation:** Measures accurate distance in meters from the Kuye river and civic amenities.
- **Automated Evacuation Routing:** Directs vulnerable settlements to safe, elevated government shelters.
- **Block Development & Vulnerability Index (BDVI):** Mathematical 0–100 score prioritizing civic interventions.

3. Field Analysis Findings (Scientific Priority Matrix)

PRIORITY	REGION / LOCALITY	RIVER DIST.	HOSPITAL DIST.	BDVI SCORE	RECOMMENDED ADMINISTRATIVE ACTION
Rank 1	Dwarka Para	324.8 m	713.5 m	**73.5 / 100**	**CRITICAL (Embankment & Flood Relief)**
Rank 2	Indas Peripheral	2429.1 m	2861.0 m	**63.0 / 100**	**MODERATE (Health Sub-Centre Needed)**
Rank 3	Shibpur Village	1506.9 m	1707.5 m	**61.2 / 100**	**MODERATE (Culvert & Road Upgrade)**
Rank 4	College Road Colony	753.5 m	731.3 m	**44.7 / 100**	**STABLE (Routine Maintenance)**
Rank 5	Hat Tala Market Area	608.2 m	310.2 m	**40.2 / 100**	**STABLE (Drainage Regular Cleaning)**

4. Evacuation Routing Scheme

- **Dwarka Para:** Safe shelter allocated at *Labpur High School Ground* (463.6 m evacuation distance).
- **Hat Tala Village:** Safe shelter allocated at *Labpur High School Ground* (781.4 m evacuation distance).
- **College Road Colony:** Safe shelter allocated at *Sambhunath College Campus* (203.4 m evacuation distance).

LABPUR 70-YEAR ENVIRONMENTAL AUDIT & 2036 FORECAST

Decadal Transformation Assessment (1950–2026) & Administrative Recommendations

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Principal Investigator: MANGAL MONDAL **Scope:** Ecological & Infrastructure Audit **Jurisdiction:** Labpur Block, Birbhum

1. Decade-Wise Historical Audit (1950 – 2026)

DECADE	GREEN CANOPY	HOUSING PATTERN	KUYE RIVER STATUS	CONNECTIVITY
1950-60	~75% canopy; dense Sal, Mahua, Babool.	90% mud-thatch housing.	Deep, perennial active river flow.	Kaccha dirt tracks, bullock carts.
1970-80	~55% canopy; cleared for agriculture.	Tin & asbestos roofs emerged.	River siltation initiated.	Morram road connectivity.
1990-00	~35% canopy; social forestry spread.	Brick & mortar town settlement (~40%).	Shallowing of river basin.	Bituminous paved roads.
2010-20	~20% canopy; old tree removal.	55%+ concrete pucca houses.	Elevated riverbed, flash inundation.	Broad-gauge railway overhaul.
2026	**~12-14% canopy**	72% concrete structures.	Severe loss of depth & drainage capacity.	Motorized EV & town bypass.

2. Net Balance Sheet: Gains vs. Losses

MAJOR DEVELOPMENTAL GAINS	CRITICAL ECOLOGICAL LOSSES
✔ All-weather road & broad-gauge railway connectivity.	✖ ~65% net reduction of indigenous tree canopy.
✔ 70%+ shift from vulnerable mud huts to pucca houses.	✖ Severe silt accumulation raising Kuye riverbed.
✔ Modernized Rural Hospital with digital health facilities.	✖ Depleting groundwater table from excessive extraction.

3. Predictive Machine Forecast: Labpur in 2036

Risk Warning (Status Quo Scenario): If unchecked deforestation continues, algorithmic projection shows:

- **Built-up Concrete Area:** Will rise to **84.2%** in core settlements.
- **Natural Tree Cover:** Will drop to an alarming **7.5%**.
- **Flood Vulnerability:** Monsoonal inundation frequency in Dwarka Para will surge by **1.8x**.

4. Strategic Recommendations for BDO Labpur

1. **Kuye River Desiltation (Dredging):** Immediate mechanical dredging of 2.5 km river stretch.
2. **Riverbank Buffer Afforestation:** Create a 100-meter native tree green belt under MGNREGS.
3. **Digital Drainage Grid:** Implement graded conduits guided by the GIS Elevation Matrix.